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$$\begin{aligned} & \frac{5-i}{2i-1} + \frac{9}{2i+1} \\ &= \frac{(5-i)(2i+1) + 9(2i-1)}{(2i-1)(2i+1)} \\ &= \frac{(5-i)(2i+1) + 18i - 9}{(2i)^2 - 1} \\ &= \frac{10i + 5 - 2i^2 - i + 18i - 9}{(2i)^2 - 1} \\ &= \frac{-2i^2 + 27i - 4}{4i^2 - 1} \\ &= \frac{2 + 27i - 4}{-4 - 1} \\ &= \frac{27i - 2}{-5} \\ &= \frac{-2}{-5} - \frac{27}{5}i \\ &= \frac{2}{5} - \frac{27}{5}i \end{aligned}$$

References